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📄 github.com/Hongsworth

📄 hongsworth.github.io/

Education

2020-2024 **MENG Electronic and Information Engineering, Imperial College London,**
London, Integrated masters

Experience

- April - October 2023 **Internship, Qualcomm**
- Worked with the modem systems team on Polar transmitter projects and developed an algorithm for zero crossing avoidance for the 16QAM modulation scheme in MATLAB.
 - Met the Bluetooth and internal requirements for performance metrics regarding EVM, ACP etc.
 - The results were presented to the wider team and the algorithm will be implemented in future Bluetooth products.
- July 2019 **Work experience, Edwards Vacuum**
- Worked with a team of 3 other students to learn more about the engineering work environment and completed team exercises.

Projects

- 2025 2D platforming game made using Unity and C#
- 2024 Final year project - Meta-learning for wireless communication optimisation using pytorch
- 2024 Built a GAN and VAE for the MNIST dataset using Pytorch
- 2024 Built and simulated a MIMO communication system with MAI and CDMA in MATLAB
- 2023 Completed an IoT project for a health monitor that takes air data to give health advice and information using Python
- 2023 Completed an embedded systems project using a STM32 to create a digital keyboard using C++ with CAN bus functionality
- 2023 Worked in a team to complete a working rover that scans for objects in an arena and communicates with a local server that displays a map containing information to the end user
- 2022 Built a decision tree using Python and used Pytorch to build a neural network to predict housing prices
- 2022 Built a simple game using the Intel DE10-lite accelerometer as an input device and interacted with a server using Amazon AWS
- 2021 Built a CPU with a subset of the MIPS ISA using SystemVerilog
- 2021 Built a fully functioning Dual-Core CPU with UART and Floating point support using ISSIE

Awards

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| 2021 | Awarded with the EE department prize for best 1st year end of year at Imperial College London |
| 2015-2020 | Won several Gold and silver awards in challenges organized by the UK Mathematical Trust and advanced to the next stage two times. |

Skills

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|--------------------|--|
| Languages | C++, C, SystemVerilog, Linux, CSS, Javascript, Reactjs, Python, Pytorch, Bash, MATLAB, SQL |
| Software and tools | Intel Quartus, SVN, Git, WSL, AWS |